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B.Tech. Degree VII Semester Supplementary Examination
April 2019

ME 15-1703 MACHINE DESIGN II
(2015 Scheme)

Time: 3 Hours

Maximum Marks: 60

PART A
(Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) What are the advantages of using a multidisc clutch?
(b) Explain the working of an internal expansion brake.
(c) Discuss the methods of adjusting the belt tensions.
(d) List the advantages of a chain drive over belt drive.
(e) How do you select a gear for an application?
(f) List the gear tooth failures and their remedies.
(g) How do you find the equivalent bearing load? Explain.
(h) What are the desirable properties of a bearing material?
(i) What are the design recommendations for rolled sections?
(j) How do you modify your design for easiness of manufacturing?



PART B

(4 × 10 = 40)

- II. (a) A multi disk clutch consists of two steel disks and one bronze disk. The inner and outer diameters of the contacting surfaces are 200 mm and 250 mm respectively. The coefficient of friction is 0.1 and maximum pressure is limited to 0.2 N/mm². Assuming uniform wear theory, calculate the required operating force and power transmitting capacity at 720 rpm. (5)
(b) A leather faced cone clutch transmits power at 600 rpm. The semi cone angle alpha is 12.5°. The mean diameter of the clutch is 300 mm while the face width of the contacting surface of the friction lining is 100 mm. The coefficient of friction is 0.23 and maximum intensity of pressure is 0.1 N/mm². Calculate the operating force and power transmitting capacity. (5)
- OR**
- III. (a) A V belt drive is required for a 15 kW, 1440 rpm, electric motor, which drives a centrifugal pump running at 360 rpm for a service of 24 hours per day. From space considerations the centre distance should be approximately 1 m. Design the belt. (5)
(b) A double block brake consists of two symmetrical pivoted shoes. The diameter of the brake drum is 300 mm and the angle of wrap for each shoe is 90°. The pivot of the shoe is located in such a way as to avoid the couple due to frictional forces. Determine the distance of the pivot from the axis of the brake drum. (5)

(P.T.O.)

- IV. A pair of spur gears consists of a 24 teeth pinion, rotating at 1000 rpm and transmitting power to a 48 teeth gear. The module is 6 mm, while the face width is 60 mm. Both the pinion and gear are made of steel with S_{ut} of 450 N/mm². They are heat treated to surface hardness of 250 BHN. The gears are machined to the accuracy of grade 8. Calculate (i) beam strength (ii) wear strength (iii) dynamic load by Spott's equation (iv) rated power that the gears can transmit if the service factor and the factor of safety are 1.5 and 2 respectively. (10)

OR

- V. The following data is given for a pair of parallel helical gears made of steel. (10)
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|------------------------------|----------------------------------|
| Speed of pinion = 720 rpm | Number of teeth on pinion = 35 |
| Number of teeth on gear = 70 | Centre distance = 285 mm |
| Normal module = 5 mm | Face width = 50 mm |
| Normal pressure angle = 20° | S_{ut} = 450 N/mm ² |
| Surface hardness = 300 BHN | Grade of machining = 8 |
| Factor of safety = 2 | Service factor = 1.25 |
- Calculate the helix angle, beam strength, wear strength, dynamic load by Spotts equation, maximum static load the gear can transmit and the power transmitting capacity.

- VI. A transmission shaft transmitting 8 kW at 400 rpm from a bevel gear G1 to helical gear G2 and mounted on two taper roller bearings B1 and B2. The gear tooth forces on the helical gear act at a pitch circle radius of 5 mm, while those on the bevel gear can be assumed to act at the large end of the tooth at a radius of 50 mm. The diameter at the bearing B1 and B2 is 40 mm. The load factor is 1.2 and expected life for 90% of the bearings is 10,000 hr. Bearing B1 and B2 are identical. Thrust force due to bevel and helical gears are taken by bearing B2. Draw the arrangement from the above data and draw the horizontal and vertical force diagrams and select suitable taper roller bearing for this application. (10)

OR

- VII. (a) A ball bearing is subjected to a radial force of 2500 N and an axial force of 1000 N. The dynamic load carrying capacity of the bearing is 7350 N. The values of X and Y factors are 0.56 and 1.6 respectively. The shaft is rotating at 720 rpm. Calculate the life of the bearing. (5)
- (b) A 360° hydrodynamic bearing operates under the following conditions: (5)
- | | |
|----------------------------------|-----------------------------------|
| Radial load = 50 kN | Journal diameter = 150 mm |
| Bearing length = 150 mm | Radial clearance = 0.15 mm |
| Minimum film thickness = 0.03 mm | Viscosity of the lubricant = 8 cP |
- What is the minimum speed of operation for the journal to work under hydrodynamic conditions?

- VIII. (a) Discuss the design recommendations for screw machine parts. (5)
- (b) Discuss the design recommendations for parts produced on milling machine. (5)

OR

- IX. (a) Explain the preparation of working drawing for manufacturing. (5)
- (b) Write short notes on tolerance and surface finish. (5)
